Independent strings in primitive groups: a bounded repair for L044

Human Compression project

An independent generating family means an irredundant one: omitting any member gives a proper subgroup. A string of involutions satisfies $[t_i, t_j] = 1$ whenever $|i - j| > 1$. No intersection property is assumed in this note.

1 The literal independent-generation claim

Proposition 3.3 on physical page8 of the supplied 2016 arXiv edition of *Highest rank of a polytope for A_n*, by Cameron, Fernandes, Leemans and Mixer, asserts an $n - 4$ bound for arbitrary independent generating families of primitive groups of degree $n \geq 8$ other than A_n, S_n. The following literal permutations contradict that statement:

$$t_0 = (2\,3)(6\,7), \qquad t_1 = (1\,4)(6\,7),$$
$$t_2 = (2\,4)(5\,7), \qquad t_3 = (2\,7)(4\,5),$$
$$t_4 = (1\,2)(3\,4)(5\,6)(7\,8).$$

The required exact certificate generates this permutation group directly, verifies that it is primitive of order1344, and obtains the five omission orders

$$192, \quad 192, \quad 192, \quad 96, \quad 168.$$

Thus these five even involutions are independent, whereas $n-4 = 4$. The group is also checked to be the actual degree-eight catalogue copy of $\mathrm{AGL}_3(2)$; that identification is not needed for the counterexample. The input is the five literal permutations, not a choice-dependent isomorphism from a different action.

This family has no string ordering: t_0 fails to commute with t_2, t_3, t_4, so its noncommuting graph has a vertex of degree three. Every string ordering would make that graph a subgraph of a path. The certificate also checks all120 orders as a confirmation. Consequently this is not a counterexample to the source's principal theorem about string C-groups.

2 The replacement needed for a projected string

Theorem 1. *Let $P \leq S_n$ be primitive, $n \geq 8$, with $P \neq A_n, S_n$. An independent generating string of involutions for P has length at most $n - 4$.*

Proof. First suppose its noncommuting diagram is disconnected. The accepted primitive commuting-factor argument applies without an intersection property: commuting nontrivial normal factors are transitive and regular, and each is simple, since any of its proper normal subgroups would be normal in the whole primitive group. Their intersection is trivial unless the group is regular abelian of prime degree, which cannot occur here with involutory generators. There are exactly two factors T, and $P = T \times T$ acts on $n = |T| \geq 60$ letters. Independence supplies a strict subgroup chain, so the length is at most $2 \log_2 n < n - 4$.

For connected diagrams, degrees8 through25 are settled by the complete finite argument below. For $n \geq 26$, retain Maróti's original primitive-order alternative, as precisely stated in the accepted L047 common proof. Its product-action case has $A_m^\ell \leq P \leq S_m \wr S_\ell$, $m \geq 5$, $1 \leq k \leq m/2$ after replacing subsets by complements, and $n = \binom{m}{k}^\ell$. If $\ell = 1$, then $k \geq 2$, and Whiston's bound in the faithful

m-point action gives length at most $m-1 \le \binom{m}{2} - 4 \le n-4$. If $\ell \ge 2$, the faithful $m\ell$-point action and Whiston give length at most $m\ell-1$. The inequality $m^\ell \ge m\ell+3$ for $m \ge 5, \ell \ge 2$ implies $m\ell-1 \le n-4$. No quotient intersection property is used.

The Mathieu actions have degree at most24. In the remaining alternative, put $q = \lfloor \log_2 n \rfloor$ and $M(n) = n\prod_{j=0}^{q-1}(n-2^j)$. The certificate checks the38 exact inequalities $M(n) < 2^{n-3}$ for $26 \le n \le 63$. For $n \ge 64$, one has

$$M(n) < n^{q+1} < 2^{(q+1)^2} \le 2^{2^q-3} \le 2^{n-3},$$

where $(q+1)^2 \le 2^q - 3$ holds at $q = 6$ and persists by induction. A family of length at least $n-3$ gives a strict subgroup chain of that length, hence $|P| \ge 2^{n-3}$, contradicting the retained order bound. This argument uses neither Conder's bound nor the intersection property. □

3 The complete finite certificate

The program visits every primitive catalogue action of degree8 through25 other than A_n, S_n. Catalogue completeness and exact GAP group operations are retained computational premises. Let d be the length of a hypothetical independent generating string. Necessarily

$$d \le \Omega(|P|), \qquad d \le 1 + \Omega(|C_P(t_0)|).$$

The first is the strict-chain bound, with prime factors counted with multiplicity. For the second, the independent subfamily $t_0, t_2, \ldots, t_{d-1}$ lies in $C_P(t_0)$ and has $d-1$ members. All involutions together must generate P. Testing these necessary conditions at $d \ge n-3$ leaves exactly

$$(8,3), \quad (8,5), \quad (9,7), \quad (16,11)$$

in the pinned current PrimGrp catalogue. Their orders are respectively $1344, 336, 432, 322560$. These current indices are not source table labels.

For each of these four actual actions, enumerate every independent connected involutory string of length from $n-3$ to $\Omega(|P|)$. At a prefix $s_1, \ldots, s_j$ generating H, extend by each involution outside H which commutes with $s_1, \ldots, s_{j-1}$ and does not commute with s_j. Reject an extended family if any member becomes redundant. Every subfamily of an independent family is independent, so this loses no admissible completion. If $[P:H] < 2^{\max(0,n-3-j)}$, the remaining strict chain is impossible. Use the full normalizer of P in S_n at the empty prefix, then the pointwise conjugation stabilizer of each fixed prefix. These are genuine symmetries of all the completion conditions; taking one representative of each candidate orbit is exhaustive up to simultaneous conjugation. A prefix already generating P cannot extend independently. *There is no intersection-property test and no centralizer-quotient pruning test.*

The complete lists at or above the forbidden ranks are empty. The numbers of visited prefixes at successive depths are

$(8,3)$	$1, 3, 18, 180, 26, 0, 0, 0, 0$
$(8,5)$	$1, 2, 11, 31, 0, 0, 0$
$(9,7)$	$1, 2, 10, 18, 2, 0, 0, 0$
$(16,11)$	$1, 4, 82, 3987, 6385, 6905, 246, 0, 0, 0, 0, 0, 0, 0, 0.$

A positive smaller-degree control recovers independent rank-three strings for the actual degree-six A_5 action. A second control checks the Coxeter string for S_4 directly against the same conditions.

Attribution and boundary. The source supplies the primitive reduction and its proposed stronger statement; this note supplies an explicit correction and the narrower replacement. The retained Maróti and Whiston theorems and complete L047 common proof remain declared inputs. The search is an adaptation of the previously accepted L047 prefix engine with all IP-specific tests removed. Its four new complete finite executions are required evidence. Earlier author roles are disclosed; old distinct acceptance is reused, not newly self-adjudicated. This proves neither every branch of L044 nor its finer source tables, and asserts no new conjecture, historical priority, or formal Lean coverage.